

Predicting Mental Health Trends

Launch Report

Group 11
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Abstract

This report outlines the goals, methodology, and evaluation strategy for predicting mental health trends using Reddit discussions. It includes work completed in DSCI 591 and proposed enhancements for DSCI 592, including severity modeling, classification of disorders, and a robust evaluation pipeline.

1 Introduction

Mental health has increasingly become a critical area of concern, particularly among young adults and students...

1.1. Summary of Work in DSCI 591

In DSCI 591, our focus was on defining the project scope, selecting appropriate data sources, and initiating exploratory data analysis. Key accomplishments included:

- **Data Collection:** We successfully collected Reddit data from mental health-related subreddits using the PRAW API. Subreddits like `r/anxiety`, `r/depression`, and `r/ADHD` were chosen based on community activity and relevance.
- **Data Preprocessing:** Raw data was cleaned by removing null entries, duplicates, and non-textual content. Text normalization techniques such as lemmatization, lowercasing, and punctuation removal were applied.
- **Initial EDA:** We performed exploratory data analysis to visualize trends in post volume, user engagement, and word frequencies. Sentiment distributions and word clouds provided insight into the emotional tone of discussions.
- **Team Setup and Planning:** We defined team roles, communication structure, and created a timeline outlining milestones for future work.

1.2. Objectives for DSCI 592

In DSCI 592, we aim to advance from data exploration to model building and evaluation. Our key objectives include:

- **Severity Modeling:** Develop a scoring mechanism to classify posts into severity levels (high, medium, low) based on contextual keywords, sentiment, and content depth.
- **Prediction of Mental Health Score:** Design models that predict mental health scores or emotional burden based on linguistic features and engagement metrics.
- **Disorder Classification:** Train machine learning models to classify posts into mental health categories (anxiety, depression, ADHD) using supervised learning.

- **Pipeline Optimization:** Create a streamlined, end-to-end pipeline that automates data collection, preprocessing, modeling, and visualization. Ensure scalability and reproducibility.

2 Highlighted Features

2.1. Severity Scoring Algorithm

Approach:

- Identify and weight keywords associated with mental health based on insights from research and reputable sources.
- Classify keywords into high, medium, and low severity categories.

Severity Levels:

- **High Severity:** Keywords linked to critical mental health issues (e.g., suicidal thoughts, panic attacks).
- **Medium Severity:** Keywords associated with moderate concerns (e.g., stress, anxiety).
- **Low Severity:** Keywords relating to mild issues (e.g., occasional sadness, minor mood fluctuations).

2.2. Sentiment Analysis Over Time

Approach:

- Analyze sentiment fluctuations over different timeframes to detect negative sentiment peaks or high severity scores.
- Identify correlations between sentiment changes and external factors (e.g., world events, holidays).

Outcome:

- Create visual dashboards to represent sentiment trends, mental health shifts, and major discussion topics.
- Use machine learning to predict future mental health trends based on historical data.

2.3. Hybrid Learning Approach

Supervised Learning:

- Classify posts based on severity levels (e.g., high, medium, low).
- Predict the mental health disorder being discussed (e.g., anxiety, depression, ADHD).
- Use **Random Forest** and **XGBoost** for robust, accurate classification and handling imbalanced datasets.
- Apply **Grid Search** and **Random Search** for hyperparameter tuning to optimize model performance.

Unsupervised Learning:

- Use **Latent Dirichlet Allocation (LDA)** for topic modeling to uncover latent themes in discussions.
- Apply clustering methods to group similar posts based on content or sentiment patterns.

2.4. Target Variables

- **Mental Health Disorder Type:** The specific disorder identified in a post (e.g., anxiety, depression).
- **Severity Level:** The severity of the condition based on the content of the post.

3 Methodology

3.1. Feature Engineering

Text Processing:

- Perform tokenization, stopwords removal, and lemmatization to prepare the text data for analysis.

Engagement Metrics:

- Extract user engagement features such as number of likes, comments, and mentions of medications to enrich the dataset.

Merged Features:

- Create merged columns containing tokenized and lemmatized versions of the original post text for better model input representation.

3.2. Severity Modeling

- Implement a rule-based scoring system that simulates severity labels based on keyword presence within posts.
- Train a **Random Forest Regressor** on these simulated severity labels.
- Include keyword modifiers and mentions of medications as features to improve regression accuracy.

3.3. Disorder Classification

- Initially label posts based on subreddit category and presence of disorder-specific keywords.
- Use **TF-IDF vectorization** for transforming text into numerical feature vectors.
- Train a **Random Forest Classifier** using the labeled dataset to classify the mental health disorder discussed (e.g., anxiety, depression, ADHD).

4 Sponsor or Proxy User

None for now.

5 Issues

- **Limited Historical Data:** The PRAW API only allows access to recent Reddit posts (10–15 days). To mitigate this, we fetch data periodically and optimize our pipeline for continuous collection.
- **Data Imbalance:** The distribution of posts across different mental health disorders and severity levels is uneven. To address this, we apply techniques such as stratified sampling and oversampling to ensure balanced model training.
- **Overfitting Risks:** There is a potential risk of overfitting if the same keyword rules used for severity simulation are reused in feature extraction. We mitigate this by separating rule-based logic from the TF-IDF modeling pipeline to prevent data leakage.

6 Collaboration with Experts

To ensure the accuracy and reliability of our mental health predictions, we are collaborating with domain experts who bring valuable knowledge and real-world experience:

- **Ashleigh Zoerb** – Licensed Mental Health Therapist with over 10 years of experience in diagnosing and treating mental health conditions.

- **Nelca Ece Yilmaz** – Holds a Master’s degree in Psychology, providing academic insights into behavioral patterns and emotional triggers in social media text.

Their guidance helps ensure that our model’s outputs align with clinical reasoning and professional assessment standards, enhancing the credibility and impact of our research.

7 Timeline for DSCI 592

Week	Tasks
Apr 8 – Apr 14	Finalize modeling objectives and begin disorder classification
Apr 15 – Apr 21	Implement severity model and evaluate both models
Apr 22 – Apr 28	Optimize pipeline and integrate visualizations
Apr 29 – May 5	Perform error analysis and feature tuning
May 6 – May 12	Prepare draft of final report and begin video presentation
May 13 – May 19	Peer review and iterate on models and report
May 20 – May 27	Final report, presentation, and submission

8 Table of Contributions

The table below identifies contributors to various sections of this document.

Section	Topic	Writing	Editing
1	Project	Everyone	Everyone
2	Team	Everyone	Everyone
3	Plan	Everyone	Everyone